

Applying fuzzy preference relation for requirements prioritization in goal oriented requirements elicitation process

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Abstract Goal oriented requirements elicitation processes are used to identify the requirements of software according to the need of stakeholders. A system may have thousands of requirements and it is difficult to prioritize such a large number of requirements when several stakeholders participate in requirements elicitation process. In literature, we identify that existing goal oriented requirements elicitation processes like KAOS, i*, AGORA do not support to prioritize the requirements when stakeholder's preferences are in the form of linguistic variable and multi-criteria decision making approach is used in requirements elicitation process. Therefore, to address this issue, we present a method for prioritization of requirements using fuzzy based approach in goal oriented requirements elicitation process by combining α -level weighted F-preference relation and extent fuzzy analytic hierarchy process in group decision making process; and then we use binary tree sort method to get the prioritized list of requirements. Finally, the utilization of the proposed approach is demonstrated with the help of an example.

Keywords Requirements elicitation · Requirements prioritization · Decision making process · Fuzzy set theory

1 Introduction

Requirements elicitation is the first sub-process of requirements engineering (RE) and involves group decision making approaches for the selection and prioritization of requirements (Sadiq and Jain 2013; Sadiq and Shahid 2009). We visualize RE as a process that includes five sub-processes namely requirements elicitation, requirements modeling, requirements analysis, requirements verification and validation, and requirements management (Sadiq and Jain 2012). Out of these sub-processes, requirements elicitation plays an important role since it has cascading effect on other sub-processes. Requirements elicitation means to identify the requirements of a product according to the need of the customers and users.

There are various methods for requirements elicitation like goal oriented requirements elicitation process (Kavakli 2002; Kaiya 2002; Anwer and Ikram 2006; Yu 1997; Zickert 2010), quality function deployment (QFD) (Chan and Wu 2007; Pai 2006) etc. In goal oriented requirements elicitation process, high level objective of an organization are decomposed into sub-goals/requirements to construct an AND/OR graph. This graph is used to find out the functional and non functional requirements of software. QFD is a requirements gathering tool for developing new or improved products to increase customers satisfaction by integrating design engineering, marketing, manufacturing and other related activities of an organization (Karsak 2004). QFD uses four matrixes namely product planning, part planning, process planning, and production planning to organize and relate pieces of data to each other (Chan and Wu 2007). These matrixes are combined to form a basis tool of QFD, called house of quality. QFD has been widely applied in various areas like transportation and communication, electronics and

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electrical utilities, food processing, textile, and construction etc. (Prasad 1998; Pai 2006).

Requirements prioritization (RP) is an important activity of requirements elicitation method because it determines the implementation order of the requirements as well as the order of importance to some stakeholders or class of stakeholders along one or more dimensions e.g., preference, business value, cost of implementation etc. (Firesmith 2004). It depends on specified requirements and on the prediction of benefit and cost of the individual requirements (Herrmann and Daneva 2008). RP provides the following benefits to the project (Firesmith 2004): (i) It improves customer satisfaction by increasing the likelihood that the customer's most important requirements are delivered first and (ii) It enables the project manager and customers to modify the project schedule to deal with the project realities of limited resources and fixed deadlines. For this reason, we mainly focus on the RP in this work.

Prioritization of requirements on the basis of several criteria creates a multi-criteria decision making (MCDM) problem. Different stakeholders participate in requirements elicitation process and they may have different preferences for the same requirements. Wang and Xiong (2011) argue that it is because of different levels of stakeholders' experience, knowledge, and understanding. In general most of the people use linguistic terminology instead of exact numbers in describing their preferences. There is a vagueness in human judgements and often difficult to represent in exact numbers (Zadeh 1975). The weights assigned to requirements by stakeholders are often described qualitatively. Fuzzy logic handles imprecise and linguistic variables mathematically (Zadeh 1965); the terms such as *low cost*, *high performance*, *good quality* etc. can be expressed in well defined manner using fuzzy logic.

Several researchers advocate the use of fuzzy logic to deal the problems related to the prioritization and decision making. Lin et al. (2007) proposed a fuzzy based decision making procedure for data warehouse system selection. Lai et al. (2008) proposed a fuzzy based method to rank the customer requirements in a competitive environment. Liu (2010) presents a group decision making method with fuzzy set theory and genetic algorithm in QFD. In a similar study, owing to the vague concepts frequently represented in decision making environments, Zhu et al. (2012) proposed a fuzzy qualitative and quantitative softgoal interdependency graphs model for non-functional requirements correlation analysis in trustworthy software.

In literature, we identify the following RP methods: (i) Numerical assignment (Voola and Babu 2012), cost benefit analysis (Herrmann and Daneva 2008), TOPSIS (Hwang and Yoon 1981), cumulative voting (Chatzipetrou et al. 2010), priority groups (Herrmann and Daneva 2008), Top-10 requirements (Herrmann and Daneva 2008; Karlsson et al.

1998), multi-attribute utility theory (Salinesi and Kornysheva 2006), weighting method (Salinesi and Kornysheva 2006; Herrmann and Daneva 2008), analytic hierarchy process (AHP) (Saaty 2008; Sadiq et al. 2009, 2010), hierarchy AHP (Karlsson Karlsson et al. 1998), minimal spanning tree (Aho et al. 1983), bubble sort (Aho et al. 1983), binary search tree (BST) (Aho et al. 1983), and Thakurta's framework for prioritization of quality requirements for inclusion in a software project (Thakurta 2012). These methods are based on the concepts of accurate measure and crisp evaluation.

Karlsson et al. (1998) investigated various RP methods; and as a result they found that AHP is the most promising method for RP. In AHP, decision maker compare all the unique pair of requirements to determine which of the two is of higher priority and to what extent (Ishizaka and Labib 2011; Saaty 2008; Triantaphyllou and Mann 1995). If there are n requirements then $n*(n-1)$ pair wise comparison are required by decision makers (DM). Karlsson et al. (1998) defined a new method for RP called, hierarchy AHP. In this method all the unique pair wise comparison are at the same level in hierarchy. They point out that hierarchy AHP possesses similar characteristics to AHP and reduces the amount of redundancy and required number of decisions.

In our earlier work (Sadiq and Jain 2013), we found that current methods for goal oriented requirements elicitation processes (Anton 1996; Anwer and Ikram 2006; Giorgini et al. 2002; Kaiya et al. 2002, 2005; Kavakli 2002; Oshiro et al. 2003; Shibaoka et al. 2007; Yamamoto and Saeki 2007; Yu 1997; Zickert 2010) do not support to prioritize the requirements when the stakeholder's preferences are in the form of linguistic variable and MCDM approach is used in requirements elicitation process. Therefore, to address this issue, we present a method for prioritization of requirements using fuzzy based approach in goal oriented requirements elicitation (PRFGORE) process by combining: (i) α -level weighted F-preference relation (ii) extent fuzzy AHP for pairwise comparisons of functional requirements (FR) and non-functional requirements (NFR) (iii) and binary sort tree method to get the prioritized list of requirements.

This paper is organized as follows: in Sect. 2, basic concepts of fuzzy sets, linguistic variable, triangular fuzzy numbers, and fuzzy preference relation are reviewed. In Sect. 3, we present fuzzy based approach for RP when MCDM approach is used in fuzzy environment. An example is provided in Sect. 4 to show how the proposed approach works under fuzzy group decision making. Finally, conclusions are drawn in Sect. 5.

2 Fuzzy set theory

In this section, we briefly review the basic concepts of fuzzy sets, linguistic variable, triangular fuzzy numbers,

and fuzzy preference relation. The *fuzzy set*, originally proposed by Zadeh (1965), is defined as follows: In a universe of discourse U_x , a fuzzy subset A of U_x is characterized by a membership function $f_A(x)$, where $f_A : U_x \rightarrow [0, 1]$ and the membership function associates with each member of x of U_x a number of $f_A(x)$ in the interval $[0, 1]$, representing the grade of membership of x in A . *Linguistic variables* are variables whose values are words or sentences in a natural or artificial language (Zadeh 1965). For example, *poor* is a linguistic variable if its values are assumed to be the fuzzy variables labelled *very poor*, *poor*, *fair*, *good*, and *very good*; rather than the numbers 0, 1, 2, 3 etc. There are several formats of fuzzy numbers, such as Triangular, Trapezoidal, Gaussian, or Sigmoid that can be used in decision making process (Mahmood et al. 2013). In practical applications, *triangular fuzzy numbers (TFNs)* are widely used to represent the approximate value range of linguistic variables. In the proposed method we adopt TFNs because of their simplicity in both concepts and computation. TFNs can be defined as follows:

Let $\mathbf{R}$ is the real line, which is viewed as a universal set of all fuzzy sub-sets. A triangular fuzzy number A is normal, convex fuzzy subset of $\mathbf{R}$, with a piece wise linear relationship function μ_A , defined by:

$$\mu_A(X) = \begin{cases} \frac{(x-a)}{(b-a)}, & a \leq x \leq b \\ \frac{(c-x)}{(c-b)}, & b \leq x \leq c \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The TFNs can be denoted by $A = (a, b, c)$ as depicted in Fig. 1. The parameters a , b , and c respectively, indicate the smallest possible value, the most promising value, and the largest possible value that describe a fuzzy event (Liu 2010; Mahmood et al. 2013). There are several operations

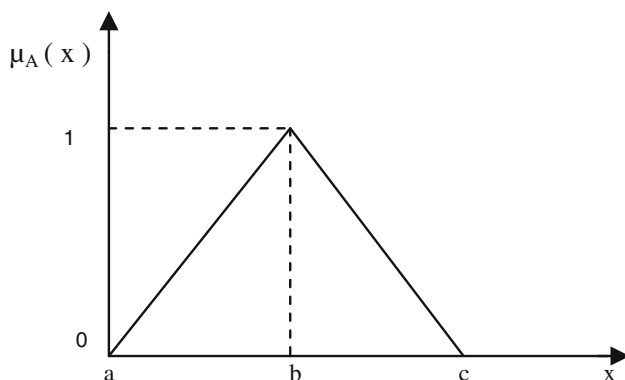


Fig. 1 The membership function of a TFN $A = (a, b, c)$

that can be performed on TFN like addition, subtraction, inverse etc.

Let $A_1 = (a_1, b_1, c_1)$ and $A_2 = (a_2, b_2, c_2)$ then:

Addition:

$$A_1 \oplus A_2 = (a_1 + a_2, b_1 + b_2, c_1 + c_2) \quad (2)$$

Subtraction:

$$A_1 \ominus A_2 = (a_1 - c_2, b_1 - b_2, c_1 - a_2) \quad (3)$$

Multiplication:

$$A_1 \odot A_2 = (a_1 \cdot a_2, b_1 \cdot b_2, c_1 \cdot c_2) \quad (4)$$

Inverse:

$$A_1^{-1} = (1/c_1, 1/b_1, 1/a_1) \quad (5)$$

Negation of

$$A_1 = (-c_1, -b_1, -a_1) \quad (6)$$

Division:

$$A_1/A_2 = (a_1/c_2, b_1/b_2, c_1/a_2) \quad (7)$$

Preference relation is a useful tool for representation of information used in decision making problems. It is used when we want to aggregate expert's preferences into group preferences. A fuzzy preference relation P on $\mathbf{R}$ is a fuzzy subset of $\mathbf{R} \times \mathbf{R}$ with membership function $f_P(A, B)$, $\forall A, B \subseteq \mathbf{R}$, where $f_P(A, B)$ represents the degree of preference of A over B (Li 1999; Lee 2005).

1. P is reciprocal if and only if $f_P(A, B) = 1 - f_P(B, A)$, $\forall A, B \subseteq \mathbf{R}$
2. P is transitive if and only if $f_P(A, B) \geq 1/2$ and $f_P(B, C) \geq 1/2 \rightarrow f_P(A, C) \geq 1/2$, $\forall A, B, C \subseteq \mathbf{R}$
3. P is a fuzzy total ordering if and only if P is reciprocal, transitive, and comparable.

In literature (Li 1999; Lee 2005) there are several interesting results on group decision making with the help of fuzzy set theory. Different aggregating methods have been used to obtain the fuzzy group preference from the fuzzy individual preferences (Li 1999; Nakamura 1986; Yuan 1991). These methods are computationally complex and difficult to implement with a usual set of judges. Therefore, in the proposed method we adopt α -level weighted F-preference relation proposed by Li (1999) because it is a simple and efficient fuzzy model associated with a solution algorithm to multiple judges; and multiple criteria decision problem are involved in a fuzzy environment, where the performance ratings and the criterion are defined imprecisely and represented by fuzzy numbers.

3 A method for prioritization of requirements using fuzzy based approach

This section presents a method for PRFGORE process. The proposed method is presented in the following:

1. Identify stakeholders and their goals.
2. Identify functional and non functional requirements. If there is a pairwise comparison among FR and NFR then apply step 4.
3. Collect DM fuzzy assessment to establish the relationship between FR and NFR and then apply step 5.
4. Apply extent fuzzy AHP for pairwise comparisons among requirements.
5. Construct comprehensive performance and weight matrix and then apply the following steps: (5.1) aggregate fuzzy performance ratings with fuzzy weights (5.2) define each sub-goal/requirements as a fuzzy number (5.3) define extended average (EA) (5.4) define the extended difference (5.5) calculate the ranking values (rv) for each requirements.
6. Apply binary tree sort method on rv of the requirements to get the prioritized list of requirements.

Step 1 Identify stakeholders and their goals

Stakeholder's identification is an important activity of requirements elicitation process. Therefore, the first step of our method is to identify the primary and secondary stakeholders (Sadiq and Jain 2013). Primary stakeholders include those who are central to any project initiative, i.e., beneficiaries, financial, politicians, sponsors, and decision maker. Secondary stakeholders include developers, experts, operators etc. Requirements analyst identifies the high level objective of the primary stakeholders and may use the following elicitation techniques: (i) interview (ii) asking 'why', 'how', and 'how-else' questions (iii) analysis of the existing documents etc. The 'why' questions are used to identify the objective of the stakeholders; how questions are used to identify lower level goals; and the how-else questions are used to find the alternatives to satisfy the higher level goals (Lamsweerde 2001).

Step 2 Identify functional and non functional requirements

In goal oriented requirements elicitation process, AND/OR graph is constructed by decomposing the high level objective of stakeholders, say G, into sub goals or requirements. This graph is used to identify the functional and non functional requirements. In AND decomposition, if all of the sub goals or requirements are achieved, their parent goals can be achieved or satisfied. On the other hand side, in OR decomposition, the achievement of at least one-

sub goals or requirements leads to the achievement of its parent goal (Lamsweerde 2001).

Step 3 Collect decision maker's fuzzy assessment

In this step, expert's opinions regarding the importance of each requirement are obtained in the form of linguistic variable such as, *very good*, *good*, *medium* etc. In this step, we collect the experts' fuzzy assessments and express their opinions on the importance of each requirement.

Step 4 Apply extent fuzzy AHP for pair wise comparisons among requirements

In this step, we adopt the extent fuzzy AHP, which was introduced by Chang (1996). Let $FR = \{fr_1, fr_2, \dots, fr_n\}$ be a set of FR and $NFR = \{nfr_1, nfr_2, \dots, nfr_m\}$ be a set of NFR. Here n equal to the number of FRs and m equals to the number of NFRs. According to the method of extent analysis (Chang 1996; Chan et al. 2013) each FR is taken and extent analysis is performed for each NFRs respectively. Therefore, the m extent analysis values for each FR are obtained as follows:

$M_{g_i}^1, M_{g_i}^2, \dots, M_{g_i}^j$, $i = 1, 2, \dots, n$, where all the $M_{g_i}^j$ ($j = 1, 2, \dots, m$) are TFNs. The value of fuzzy synthetic extent with respect to the i th FR is defined as:

$$S_i = \sum_{j=1}^m M_{g_i}^j \odot \left[\sum_{i=1}^n \sum_{j=1}^m M_{g_i}^j \right]^{-1} \quad (8)$$

To obtain $\sum_{j=1}^m M_{g_i}^j$, the fuzzy addition operation of m extent analysis values for a particular matrix is performed as:

$$\sum_{j=1}^m M_{g_i}^j = \left(\sum_{j=1}^m a_j, \sum_{j=1}^m b_j, \sum_{j=1}^m c_j \right) \quad (9)$$

and to obtain $\left[\sum_{i=1}^n \sum_{j=1}^m M_{g_i}^j \right]^{-1}$, the fuzzy addition operation of $M_{g_i}^j$ ($j = 1, 2, \dots, m$) values is performed such as:

$$\sum_{i=1}^n \sum_{j=1}^m M_{g_i}^j = \left(\sum_{i=1}^n a_i, \sum_{i=1}^n b_i, \sum_{i=1}^n c_i \right) \quad (10)$$

and then the inverse of the above vector can be calculated as:

$$\left[\sum_{i=1}^n \sum_{j=1}^m M_{g_i}^j \right]^{-1} = \left[1 / \sum_{i=1}^n c_i, 1 / \sum_{i=1}^n b_i, 1 / \sum_{i=1}^n a_i \right] \quad (11)$$

Let $A_1 = (a_1, b_1, c_1)$ and $A_2 = (a_2, b_2, c_2)$ are two TFNs. The degree of possibility of A_1 and A_2 is defined as follows:

$$V(A_1 \geq A_2) = \sup_{x \geq y} [\min(\mu_{A_1}(x), \mu_{A_2}(y))] \quad (12)$$

When a pair (x, y) exists such that $x \geq y$ and $\mu_{A_1}(x) = \mu_{A_2}(y) = 1$, then we have $V(A_1 \geq A_2) = 1$. Since A_1 and A_2 are convex fuzzy numbers, we have that:

$$V(A_1 \geq A_2) = 1 \text{ iff } b_1 \geq b_2, \quad (13)$$

$$V(A_2 \geq A_1) = \text{hgt}(A_1 \cap A_2) = \mu_{A_1}(d) \quad (14)$$

where d is the ordinate of the highest intersection point D between μ_{A_1} and μ_{A_2} (see Fig. 2). When $A_1 = (a_1, b_1, c_1)$ and $A_2 = (a_2, b_2, c_2)$, the ordinate of D is given by:

$$V(A_1 \geq A_2) = \text{hgt}(A_1 \cap A_2) = \frac{(a_1 - c_1)}{(b_2 - c_2) - (b_1 - a_1)} \quad (15)$$

To compare A_1 and A_2 , both the values of $V(A_1 \geq A_2)$ and $V(A_2 \geq A_1)$ are required. The degree possibility for a convex fuzzy number to be greater than k convex fuzzy numbers A_i ($i = 1, 2, \dots, k$) can be defined by

$$\begin{aligned} V(A \geq A_1, A_2, \dots, A_k) &= V[(A \geq A_1) \\ &\text{and } (A \geq A_2) \text{ and } \dots \text{ and } (A \geq A_k)] \\ &= \min V(A \geq A_i), i = 1, 2, \dots, k. \end{aligned} \quad (16)$$

$$\text{If } d(X_i) = \min V(S_i \geq S_k) \text{ for } k = 1, 2, \dots, n, k \neq i \quad (17)$$

Then the weight vector (WV) is given by:

$$WV' = (d'(X_1), d'(X_2), \dots, d'(X_n))^T$$

where X_i ($i = 1, 2, \dots, n$) are n FR. Via normalization, the normalized WV:

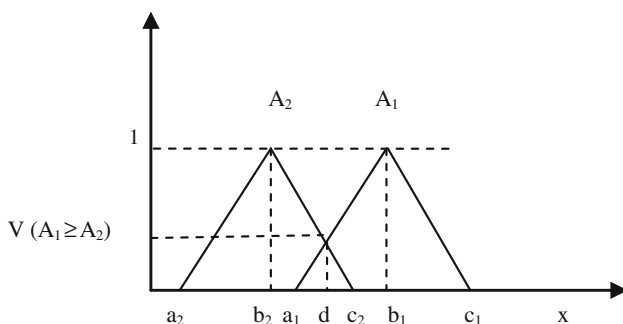


Fig. 2 Intersection between two fuzzy numbers

$$WV = (d(X_1), d(X_2), \dots, d(X_n))^T \quad (18)$$

Where WV is a non fuzzy number?

Step 5 Compute fuzzy group preference from the fuzzy individual preferences

For the prioritization of requirements on the basis of various criteria's, we aggregate fuzzy performance rating through all decision maker by means of extended addition and scalar multiplication to form a comprehensive performance matrix P , in which performance rating:

$$p_{ij} = (1/n) \odot (P_{ij}^1 \oplus P_{ij}^2 \oplus \dots \oplus P_{ij}^n) \quad (19)$$

is a triangular fuzzy number of the form:

$$(p_{1ij}, p_{2ij}, p_{3ij}) = \left(\frac{1}{n} \sum_{k=1}^n p_{1ij}^k, \frac{1}{n} \sum_{k=1}^n p_{2ij}^k, \frac{1}{n} \sum_{k=1}^n p_{3ij}^k \right) \quad (20)$$

Now calculate the fuzzy weight through all DM by means of extended addition and scalar multiplication to form a comprehensive WV. Once we have obtained the comprehensive performance and weight matrix then apply the following steps (Li 1999):

Step 5.1 Aggregate fuzzy ratings with fuzzy weights by means of extended multiplication to form a weighted, comprehensive decision matrix D , in which

$$d_{ij} = p_{ij} \odot w_j \quad (21)$$

is a fuzzy number with parabolic membership functions in the form of:

$$(\lambda_{1ij}, \lambda_{2ij}, \lambda_{3ij}/d_{ij}/\Delta_{1ij}, \Delta_{2ij}, \Delta_{3ij});$$

where

$$\lambda_{1ij} = (w_{2j} - w_{1j})(p_{2ij} - p_{1ij}) \quad (22)$$

$$\lambda_{2ij} = w_{1j}(p_{2ij} - p_{1ij}) + p_{1ij}(w_{2j} - w_{1j}) \quad (23)$$

$$\lambda_{3ij} = w_{1j}p_{1ij} \quad (24)$$

$$\Delta_{1ij} = (w_{3j} - w_{2j})(p_{3ij} - p_{2ij}) \quad (25)$$

$$\Delta_{2ij} = w_{3j}(p_{3ij} - p_{2ij}) + p_{2ij}(w_{3j} - w_{2j}) \quad (26)$$

$$\Delta_{3ij} = w_{3j}p_{2ij} \quad (27)$$

and

$$d_{ij} = w_{2j}p_{2ij} \quad (28)$$

Step 5.2 Define each sub-goal/requirement as a fuzzy number A_i , $i = 1, 2, \dots, m$ by means of extended

addition and scalar multiplication through the following criteria:

$$A_i = 1/m \odot (d_{i1} \oplus d_{i2} \oplus \dots \oplus d_{ic}) \quad (29)$$

with parabolic membership function in the form of: $(\lambda_{1i}, \lambda_{2i}, \lambda_{3i}/EA_i/\Delta_{1i}, \Delta_{2i}, \Delta_{3i})$ where

$$\lambda_{li} = \frac{1}{m} \sum_{j=1}^m \lambda_{lij}, I = 1, 2, 3; \quad (30)$$

$$\Delta_{li} = \frac{1}{m} \sum_{j=1}^m \Delta_{lij}, I = 1, 2, 3, \quad (31)$$

and

$$EA_i = \frac{1}{m} \sum_{j=1}^m d_{ij}. \quad (32)$$

Step 5.3 Define EA by means of extended addition and scalar multiplication through all alternatives (sub-goals/requirements).

$$EA = 1/n \odot (g_1 \oplus g_2 \oplus \dots \oplus g_h) \quad (33)$$

with parabolic membership function in the form of: $(\lambda_1, \lambda_2, \lambda_3/\text{Sum_EA}/\Delta_1, \Delta_2, \Delta_3)$; where

$$\lambda_i = \frac{1}{n} \sum_{ij=1}^n \lambda_{li}, I = 1, 2, 3; \quad (34)$$

$$\Delta_i = \frac{1}{n} \sum_{i=1}^n \Delta_{li}, I = 1, 2, 3, \quad (35)$$

$$\text{Sum_EA} = \frac{1}{n} \sum_{i=1}^n EA_i \quad (36)$$

Step 5.4 Define the extended difference, $EA_i \ominus \text{Sum_EA}$, for each $A_i \in R$, with parabolic membership function in the form of: $((\lambda_{1i} - \Delta_1), (\lambda_{2i} + \Delta_2), (\lambda_{3i} - \Delta_3)/EA_i - \text{Sum_EA}/(\Delta_{1i} - \lambda_1), (-\Delta_{2i} - \lambda_2), (\Delta_{3i} - \lambda_3))$.

Step 5.5 Calculate rv of each requirements

In this step, we calculate the ranking values (rv_i) for each requirements A_i by means of F-preference relation R : if $(\lambda_{3i} - \Delta_3) < 0, (\Delta_{3i} - \lambda_3) \geq 0, EA_i \geq \text{Sum_EA}$; then

$$rv_i = \mu R(A_i \ominus EA, 0) = \Pi^+ / (\Pi^+ + \Pi^-); \quad (37)$$

else if $(\lambda_{3i} - \Delta_3) \leq 0, (\Delta_{3i} - \lambda_3) > 0, EA_i \leq \text{Sum_EA}$; then

$$rv_i = \mu R(A_i \ominus EA, 0) = \Psi^+ / (\Psi^+ + \Psi^-); \quad (38)$$

else if $(\lambda_{3i} - \Delta_3) = 0, (\Delta_{3i} - \lambda_3) = 0, EA_i = \text{Sum_EA}$ then

$$rv_i = \mu R(A_i \ominus EA, 0) = 0.5; \quad (39)$$

else if $(\lambda_{3i} - \Delta_3) \geq 0, (\Delta_{3i} - \lambda_3) > 0, EA_i \geq \text{Sum_EA}$ then

$$rv_i = \mu R(A_i \ominus EA, 0) = 1; \quad (40)$$

else if $(\lambda_{3i} - \Delta_3) < 0, (\Delta_{3i} - \lambda_3) \leq 0, EA_i \leq \text{Sum_EA}$ then

$$rv_i = \mu R(A_i \ominus EA, 0) = 0. \quad (41)$$

where

$$\begin{aligned} \Pi^+ = & -[1/4(\Delta_{1i} - \lambda_1) - 1/3(\Delta_{2i} + \lambda_2) + 1/2(\Delta_{3i} - \lambda_3)] \\ & + [1/4(\lambda_{1i} - \Delta_1)(1 - Z^4) + 1/3(\lambda_{2i} + \Delta_2)(1 \\ & - Z^3) + 1/2(\lambda_{3i} - \Delta_3)(1 - Z^2)]; \end{aligned} \quad (42)$$

$$\begin{aligned} \Pi^- = & (1/4(\lambda_{1i} - \Delta_1)Z^4 + 1/3(\lambda_{2i} + \Delta_2)Z^3 + 1/2(\lambda_{3i} \\ & - \Delta_3)Z^2)]; \end{aligned} \quad (43)$$

$$Z = \left[\frac{-(\lambda_{2i} + \Delta_2) + \text{sqrt}\{(\lambda_{2i} + \Delta_2)^2 - 4(\lambda_{1i} - \Delta_1)(\lambda_{3i} - \Delta_3)\}}{2(\lambda_{1i} - \Delta_1)} \right] \quad (44)$$

$$\begin{aligned} \Psi^+ = & 1/4(\Delta_{1i} - \lambda_1)P^4 + 1/3(-\Delta_{2i} - \lambda_2)P^3 \\ & + 1/2(\Delta_{3i} - \lambda_3)P^2; \end{aligned} \quad (45)$$

$$\begin{aligned} \Psi^- = & -1/4(\lambda_{1i} - \Delta_1) + 1/3(\lambda_{2i} + \Delta_2) + 1/2(\lambda_{3i} \\ & - \Delta_3) - [1/4(\Delta_{1i} - \lambda_1)(1 - P^4) - 1/3(\Delta_{2i} + \lambda_2)(1 \\ & - P^3) + 1/2(\Delta_{3i} - \lambda_3)(1 - P^2)]; \end{aligned} \quad (46)$$

$$P = \left[\frac{(\Delta_{2i} + \lambda_2) - \text{sqrt}\{(-\Delta_{2i} - \lambda_2)^2 - 4(\Delta_{1i} - \lambda_1)(\Delta_{3i} - \lambda_3)\}}{2(\Delta_{1i} - \lambda_1)} \right]; \quad (47)$$

Where “sqrt” is a square root function?

Step 6 Binary tree sort method

In this step, we create the BST of the rv that we have obtained in previous step and then the tree is traversed in IN-ORDER. The IN-ORDER traversal of BST lists the elements in ascending order. The algorithm to traverse a non empty binary tree in IN-ORDER is given below (Aho et al. 1983):

- Traverse the left sub-tree in IN-ORDER.
- Visit the root node.
- Traverse the right sub-tree in IN-ORDER.

4 An example

The objective of this section is to apply the proposed method for RP of Institute Examination System (IES). This system is used to provide the facility to submit online examination form and generate the results of the students; and display the news related to examinations. After submitting the examination form, the system will generate the examination fee receipt and students will be able to take the printout of that receipt. The system will also generate the patterns of the sitting arrangement. RP plays a key role in requirements elicitation process to decide which requirements to implement first according to the budget, time, cost, customer expectations etc. To illustrate the proposed approach, we assume that ten stakeholders i.e., $S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8, S_9$, and S_{10} ; and five DM, i.e. DM_1, DM_2, DM_3, DM_4 , and DM_5 are participating in group requirements elicitation and analysis process.

Let S_1 is the primary stakeholder and its objective is to develop software for IES. S_1 is also the beneficiary and sponsor of the system, who wants to develop a system. S_2 is the secondary stakeholder and is responsible for the identification of FR and to deal with the decisions that are related to the organization goals and objective. The objective of stakeholder S_3 is to elicit the NFR and also to take the decisions related to the organizations goals and objective. Stakeholder S_4 is the in charge of the system throughout all the life cycle phases. Stakeholder S_5 is requirements analyst and developer; and is directly involved in information system development. S_6 is operator and deals with assuring effectiveness when performing operation within the organization. S_7 is both primary and secondary stakeholders and will act as an expert and the decision maker. S_8 is the regulators and is responsible for quality, security and cost or other aspects of the system. S_9 and S_{10} are the financers; and they benefit indirectly from the system, obtaining financial rewards and are selected on the basis of geographical location. Let the objective of S_1 is represented by G.

Let us assume that during group requirements elicitation process S_5 collect the requirements from various stakeholders and construct the AND/OR graph by decomposing the objective of S_1 into three FR, i.e., student module, administrative module, and teachers module; and there is one non-functional requirement, i.e., Trustworthiness. Trustworthiness (Zhu et al. 2012) means that the system should have one or more of the following characteristics: safety, security, performance etc. (Fukushima and Yamada 2010; Chung et al. 2000; Mairiza and Zowghi 2011). In Fig. 3, Student module, administrative module, and teacher's modules are represented by FR_1, FR_2 , and FR_3 respectively; and trustworthiness is represented by NFR.

There is an AND decomposition among FR_1, FR_2, FR_3 and NFR as shown in Fig. 3. FR_1, FR_2 , and FR_3 are further decomposed and refined into ten FRs, i.e., $fr_1, fr_2, \dots, fr_{10}$. In this study, we prioritize $fr_1, fr_2, \dots, fr_{10}$ on the basis of the nfr_1, nfr_2 , and nfr_3 . The meaning of these requirements is given below:

- (i) fr_1 : printout of bank receipt of students fee;
- (ii) fr_2 : entry of internal and external marks;
- (iii) fr_3 : view semester result;
- (iv) fr_4 : generate examination sitting arrangement;
- (v) fr_5 : online conduct of examination;
- (vi) fr_6 : fill examination form; and after successful submission of the form system will generate the following information: (a) roll number, (b) name of the students, (c) examination name, (d) subject code, (e) subject name(s), (f) number of backlogs, if any (g) examination fee(s);
- (vii) fr_7 : upload any exam related activities;
- (viii) fr_8 : generate examination hall ticket;
- (ix) fr_9 : approve examination form;
- (x) fr_{10} : on line payment of examination fee.

Student module (FR_1) is decomposed into three sub requirements, i.e., fr_1, fr_6 , and fr_{10} ; and there is an AND decomposition among these requirements. Administrative module (FR_2) is decomposed into three sub-requirements, i.e., fr_7, fr_8, fr_9 ; and there is also an AND decomposition among these requirements. FR_3 is decomposed into four sub-requirements, i.e., fr_2, fr_3, fr_4 , and fr_5 . Similarly, Trustworthiness (NFR) is further decomposed into three sub-requirements, i.e., nfr_1 : Security; nfr_2 : Reliability; and nfr_3 : Performance. There is also an AND decomposition among these requirements. Reliability, i.e., nfr_2 is further decomposed into three sub- requirements, i.e., nfr_{2-1} : recoverability; nfr_{2-2} : Adaptability; and nfr_{2-3} : maturity. There is an OR decomposition among these requirements. OR decomposition means that the selection of any requirements leads to the achievement of the parent requirements. There is a pairwise comparison among nfr_{2-1}, nfr_{2-2} , and nfr_{2-3} . Therefore, we use extent fuzzy AHP for pair wise comparisons. Let S_5 decides that nfr_{2-1} is more important than nfr_{2-2} . Therefore, according to Saaty rating scale (Saaty 2008), in DM_1 matrix, cell nfr_{2-1} and nfr_{2-2} contains 5; and cell nfr_{2-2} and nfr_{2-1} contains 1/5. S_5 also decides that nfr_{2-1} is important than nfr_{2-3} , giving 3 in cell nfr_{2-1} and nfr_{2-3} ; and 1/3 in cell nfr_{2-3} and nfr_{2-1} . This process is applied to construct all the matrices of DM, i.e., $DM_1, DM_2, \dots, DM_5$. Five DMs pairwise comparisons for nfr_{2-1}, nfr_{2-2} , and nfr_{2-3} are given as follows; and comprehensive pairwise comparisons matrix is given in Table 1.

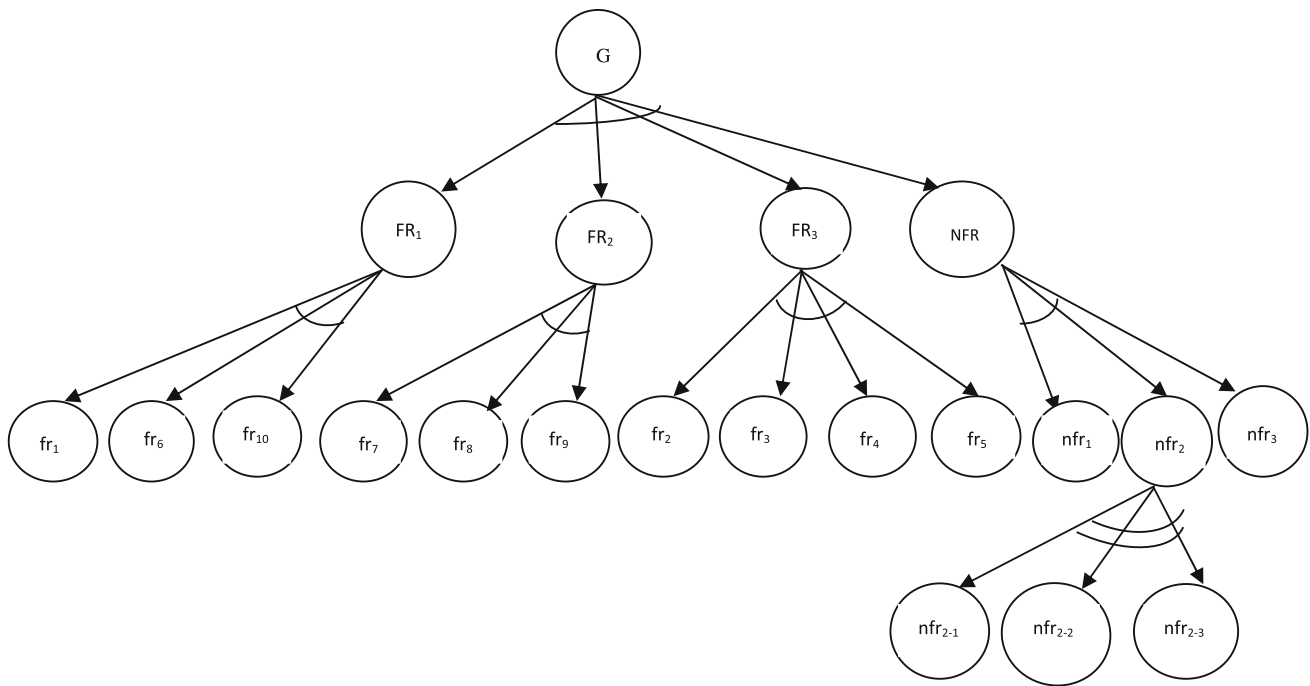


Fig. 3 AND/OR graph

$$DM_1 = \begin{matrix} & nfr_{2-1} & nfr_{2-2} & nfr_{2-3} \\ \begin{matrix} nfr_{2-1} \\ nfr_{2-2} \\ nfr_{2-3} \end{matrix} & \begin{bmatrix} 1 & 5 & 3 \\ 1/5 & 1 & 3 \\ 1/3 & 1/3 & 1 \end{bmatrix} \end{matrix}$$

$$DM_2 = \begin{matrix} & nfr_{2-1} & nfr_{2-2} & nfr_{2-3} \\ \begin{matrix} nfr_{2-1} \\ nfr_{2-2} \\ nfr_{2-3} \end{matrix} & \begin{bmatrix} 1 & 3 & 3 \\ 1/3 & 1 & 5 \\ 1/3 & 1/5 & 1 \end{bmatrix} \end{matrix}$$

$$DM_3 = \begin{matrix} & nfr_{2-1} & nfr_{2-2} & nfr_{2-3} \\ \begin{matrix} nfr_{2-1} \\ nfr_{2-2} \\ nfr_{2-3} \end{matrix} & \begin{bmatrix} 1 & 1/3 & 5 \\ 3 & 1 & 3 \\ 1/5 & 1/3 & 1 \end{bmatrix} \end{matrix}$$

$$DM_4 = \begin{matrix} & nfr_{2-1} & nfr_{2-2} & nfr_{2-3} \\ \begin{matrix} nfr_{2-1} \\ nfr_{2-2} \\ nfr_{2-3} \end{matrix} & \begin{bmatrix} 1 & 1 & 3 \\ 1 & 1 & 1 \\ 1/3 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$DM_5 = \begin{matrix} & nfr_{2-1} & nfr_{2-2} & nfr_{2-3} \\ \begin{matrix} nfr_{2-1} \\ nfr_{2-2} \\ nfr_{2-3} \end{matrix} & \begin{bmatrix} 1 & 3 & 1 \\ 1/3 & 1 & 5 \\ 1 & 1/5 & 1 \end{bmatrix} \end{matrix}$$

By applying formula (8), we have

$$S_{nfr_{2-1}} = (2.3, 8.47, 11) \odot (1/23, 1/15.68, 1/5.9) = (0.099, 0.542, 1.859)$$

$$S_{nfr_{2-2}} = (2.2, 5.393, 9) \odot (1/23, 1/15.68, 1/5.9) = (0.095, 0.345, 1.521)$$

$$S_{nfr_{2-3}} = (1.43, 1.82, 3) \odot (1/23, 1/15.68, 1/5.9) = (0.060, 0.116, 0.507)$$

By applying formulae (15) and (16), we have the following values:

$$V(S_{nfr_{2-1}} > S_{nfr_{2-2}}) = 1$$

$$V(S_{nfr_{2-1}} > S_{nfr_{2-3}}) = 1$$

$$V(S_{nfr_{2-2}} > S_{nfr_{2-1}}) = 0.878$$

$$V(S_{nfr_{2-2}} > S_{nfr_{2-3}}) = 1.186$$

$$V(S_{nfr_{2-3}} > S_{nfr_{2-1}}) = 0.489$$

$$V(S_{nfr_{2-3}} > S_{nfr_{2-2}}) = 0.643$$

$$d'(nfr_{2-1}) = \min(1, 1) = 1$$

$$d'(nfr_{2-2}) = \min(0.878, 1.186) = 0.878$$

$$d'(nfr_{2-3}) = \min(0.489, 0.643) = 0.489$$

After applying formula (17), we obtained the value of priority weight vector $(WV') = (1, 0.878, 0.489)$. After normalization we obtained the following WVs with respect to the decision criteria nfr_{2-1} , nfr_{2-2} , and nfr_{2-3} .

$$WV = (0.422, 0.371, 0.207)$$

Table 1 Comprehensive pairwise comparisons matrix

	nfr ₂₋₁	nfr ₂₋₂	nfr ₂₋₃
nfr ₂₋₁	(1,1,1)	(0.3,2.47,5)	(1,5,5)
nfr ₂₋₂	(0.2,0.993,3)	(1,1,1)	(1,3,4,5)
nfr ₂₋₃	(0.2,0.42,1)	(0.2,0.4,1)	(1,1,1)

Table 2 Triangular fuzzy numbers of linguistic values for each FR

S. no.	Linguistic values	Triangular fuzzy number
1	VL (Very low)	(0,0,0.25)
2	L (Low)	(0,0.25,0.5)
3	M (Middle)	(0.25,0.5,0.75)
4	H (High)	(0.5,0.75,1)
5	VH (Very high)	(0.75,1,1)

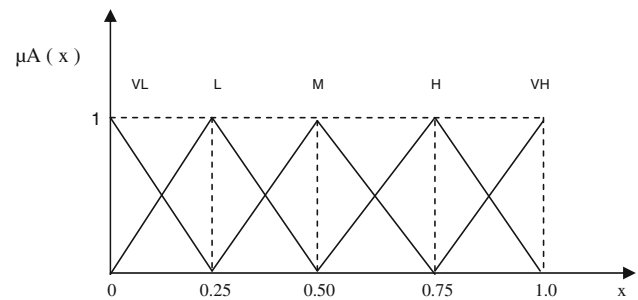
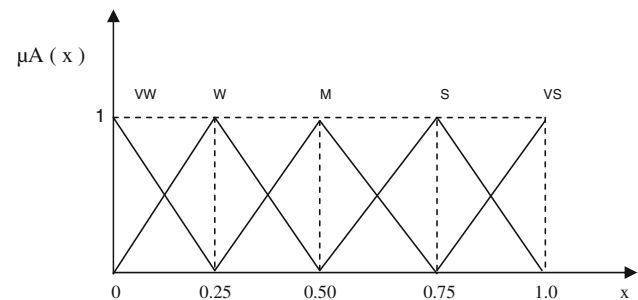
As a result, we found that nfr₂₋₁, i.e., recoverability has the highest weight value. Therefore, this requirement would be used as one of the criteria for the prioritization of ten FR, i.e., fr₁, fr₂, ..., fr₁₀. Once we have identified the criteria on which fr₁, fr₂, ..., fr₁₀ would be prioritized, then the next step is to define the linguistic values of the importance weight of each FR; and the relationship between FR and NFR. In our example, five ranks are used to evaluate the importance weight of each requirements i.e., *Very Low (VL)*, *Low (L)*, *Middle (M)*, *High (H)*, and *Very High (VH)*; and five ranks are used to evaluate the relationship between functional and non functional requirements, i.e., *Very Weak (VW)*, *Weak (W)*, *Medium (M)*, *Strong (S)*, and *Very Strong (VS)*.

The TFN of linguistic values for each FR and the relationship between FR and NFR are listed in Tables 2 and 3 respectively. The membership function of TFNs in Tables 2 and 3 are displayed in Figs. 4, 5, respectively.

The relationship between functional and non functional requirements is evaluated by cross functional team with five DM (see Table 4). Ten stakeholders were asked to evaluate the importance of each NFR; and the fuzzy importance weight of these stakeholders for each NFR is given in Table 5. Now we aggregate fuzzy importance weight for each NFR by applying formula (20). Then we aggregate the fuzzy relationship between functional and

Table 3 Triangular fuzzy numbers of linguistic values for the relationship between FR and NFR

S. no.	Linguistic values	Triangular fuzzy number
1	VW (Very weak)	(2,2,4)
2	W (Weak)	(2,4,6)
3	M (Medium)	(4,6,8)
4	S (Strong)	(6,8,10)
5	VS (Very strong)	(8,10,10)

**Fig. 4** Membership function of linguistic values (VL, L, M, H, VH) for each FRs**Fig. 5** Membership function of linguistic values (VW, W, M, S, VS) for the relationship between FRs and NFRs

non functional requirements; and finally the results are given in Tables 6, 7 respectively.

A weighted comprehensive decision matrix, depicted in Table 8, can be established by applying the extended multiplication discussed in step 5.1 (see formula (21)), where QF denotes the quadratic membership function of the form of:

Table 4 Relationship between FRs and NFRs evaluated by five DM

DM	Functional requirements (fr)	Non functional requirements (nfrs)			Functional requirements (fr)	Non functional requirements (NFRs)		
		nfr ₁	nfr ₂	nfr ₃		nfr ₁	nfr ₂	nfr ₃
DM ₁	fr ₁	W	S	VS	fr ₆	S	S	VS
DM ₂		S	VS	VW		VS	VS	W
DM ₃		M	S	S		S	S	S
DM ₄		S	VS	W		M	W	M
DM ₅		VS	S	S		S	VS	M
DM ₁	fr ₂	S	VS	W	fr ₇	VW	M	S
DM ₂		VS	S	VS		S	VS	S
DM ₃		W	M	S		M	S	VW
DM ₄		S	VW	VW		M	M	W
DM ₅		M	W	S		M	VS	S
DM ₁	fr ₃	VS	S	VW	fr ₈	S	VS	M
DM ₂		W	W	S		S	W	S
DM ₃		M	VS	VS		W	S	W
DM ₄		VS	M	M		S	W	S
DM ₅		VW	VW	S		S	M	S
DM ₁	fr ₄	S	S	M	fr ₉	VS	S	M
DM ₂		VS	M	M		M	M	S
DM ₃		W	VS	S		W	VW	VS
DM ₄		S	W	VS		VS	VS	S
DM ₅		S	M	S		S	M	S
DM ₁	fr ₅	VW	M	S	fr ₁₀	S	S	W
DM ₂		W	S	M		S	VS	S
DM ₃		M	W	W		VS	W	VS
DM ₄		S	VW	VW		W	S	S
DM ₅		M	S	S		VW	VS	VS

Table 5 Fuzzy importance weight by ten stakeholders for each NFR

Non functional requirements (NFRs)	Stakeholders									
	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇	S ₈	S ₉	S ₁₀
nfr ₁	VH	H	VH	H	M	L	M	VL	H	L
nfr ₂	H	L	VH	VH	H	M	L	M	M	VL
nfr ₃	M	VH	L	H	VH	H	VH	L	VL	M

Table 6 Fuzzy importance weight matrix

Non functional requirements (nfr)	Fuzzy importance weight
nfr ₁	(0.35,0.58,0.78)
nfr ₂	(0.33,0.55,0.75)
nfr ₃	(0.38,0.6,0.78)

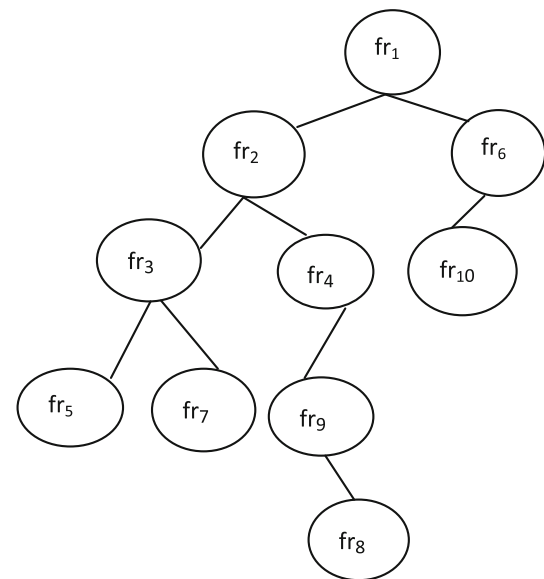
Table 7 Fuzzy relationship between FR and NFRs

Functional requirements (fr)	nfr ₁	nfr ₂	nfr ₃
fr ₁	(5.2,7.2,8.8)	(6.8,8.8,10)	(4.8,6.4,8)
fr ₂	(5.2,7.2,8.8)	(4.4,6,7.6)	(4.8,6.4,8)
fr ₃	(4.8,6.4,7.6)	(4.4,6,7.6)	(5.2,6.8,8.4)
fr ₄	(5.6,7.6,9.2)	(4.8,6.8,8.4)	(5.6,7.6,9.2)
fr ₅	(3.6,5.2,7.2)	(4.5,6,7.6)	(4.5,6,7.6)
fr ₆	(6.4,8.4,9.6)	(6.8,9.2)	(4.8,6.8,8.4)
fr ₇	(4.5,6,7.6)	(6.8,9.2)	(4.4,6,8)
fr ₈	(5.2,7.2,9.2)	(4.4,6.4,8)	(4.8,6.8,8.8)
fr ₉	(5.6,7.6,8.8)	(4.8,6.4,8)	(6.8,9.6)
fr ₁₀	(4.8,6.4,8)	(6.8,9.2)	(6.8,9.2)

Table 8 Quadratic membership functions

SRs	Non-functional requirements		
	nfr ₁	nfr ₂	nfr ₃
fr ₁	4.176 _{QF}	4.84 _{QF}	3.84 _{QF}
fr ₂	4.176 _{QF}	3.3 _{QF}	3.84 _{QF}
fr ₃	3.712 _{QF}	3.3 _{QF}	4.08 _{QF}
fr ₄	4.408 _{QF}	3.74 _{QF}	4.56 _{QF}
fr ₅	3.02 _{QF}	3.08 _{QF}	3.36 _{QF}
fr ₆	4.87 _{QF}	4.40 _{QF}	4.08 _{QF}
fr ₇	3.25 _{QF}	4.40 _{QF}	3.6 _{QF}
fr ₈	4.18 _{QF}	3.52 _{QF}	4.08 _{QF}
fr ₉	4.408 _{QF}	3.52 _{QF}	4.8 _{QF}
fr ₁₀	3.712 _{QF}	4.40 _{QF}	4.8 _{QF}

$4.176_{QF} = 0.46, 1.896, 1.82/4.176/0.32, 3.008, 6.86$
 $4.84_{QF} = 0.44, 2.156, 2.24/4.84/0.24, 2.90, 7.50$
 $3.84_{QF} = 0.352, 1.664, 1.824/3.84/0.288, 2.687, 6.24$
 $4.176_{QF} = 0.46, 1.896, 1.82/4.176/0.32, 3.008, 6.864$
 $3.3_{QF} = 0.352, 1.496, 1.452/3.3/0.32, 2.72, 5.7$
 $3.84_{QF} = 0.352, 1.664, 1.824/3.84/0.29, 2.69, 6.24$
 $3.712_{QF} = 0.368, 1.664, 1.68/3.712/0.24, 2.46, 5.93$
 $3.3_{QF} = 0.352, 1.496, 1.452/3.3/0.32, 2.72, 5.7$
 $4.08_{QF} = 0.352, 1.752, 1.98/4.08/0.288, 2.76, 6.55$
 $4.408_{QF} = 0.46, 1.98, 1.96/4.408/0.32, 3.09, 7.176$
 $3.74_{QF} = 0.44, 1.716, 1.58/3.74/0.32, 2.88, 6.3$
 $4.56_{QF} = 0.44, 1.992, 2.13/4.56/0.29, 2.9, 7.18$
 $3.02_{QF} = 0.36, 1.388, 1.26/3.02/0.40, 3.0, 5.62$
 $3.08_{QF} = 0.35, 1.408, 1.32/3.08/0.4, 3.02, 5.7$
 $3.36_{QF} = 0.352, 1.488, 1.52/3.36/0.36, 2.93, 5.93$
 $4.87_{QF} = 0.46, 2.172, 2.24/4.87/0.24, 2.86, 7.49$
 $4.40_{QF} = 0.44, 1.98, 1.98/4.4/0.24, 2.74, 6.9$
 $4.08_{QF} = 0.44, 1.816, 1.82/4.08/0.29, 2.75, 6.55$
 $3.25_{QF} = 0.37, 1.48, 1.40/3.25/0.4, 3.08, 5.93$
 $4.40_{QF} = 0.44, 1.98, 1.98/4.4/0.24, 2.74, 6.9$
 $3.6_{QF} = 0.35, 1.58, 1.67/3.6/0.36, 3, 6.24$
 $4.18_{QF} = 0.46, 1.896, 1.82/4.18/0.4, 3.4, 7.18$
 $3.52_{QF} = 0.44, 1.63, 1.45/3.52/0.32, 2.8, 6.0$
 $4.08_{QF} = 0.44, 1.82, 1.82/4.08/0.36, 3.14, 6.86$
 $4.408_{QF} = 0.46, 1.98, 1.96/4.408/0.24, 2.69, 6.86$
 $3.52_{QF} = 0.352, 1.58, 1.58/3.52, 2.80, 6.00$
 $4.80_{QF} = 0.44, 2.08, 2.28/4.80/0.288, 2.97, 7.49$
 $3.712_{QF} = 0.37, 1.67, 1.68/3.712/0.32, 2.85, 6.24$
 $4.4_{QF} = 0.44, 1.98, 1.98/4.4/0.24, 2.74, 6.9$
 $4.8_{QF} = 0.44, 2.08, 2.28/4.8/0.22, 2.59, 7.17$

**Fig. 6** Binary search tree

After applying the steps 5.2 and 5.3, the EA of all FR by means of extended addition and scalar multiplication is in the form of: $3.98_{QF} = 0.406, 1.78, 1.79/3.98/0.306, 2.86, 6.54$. After executing steps 5.4 and 5.5, we get the following rv of FR: $fr_1 = 0.587$, $fr_2 = 0.50$, $fr_3 = 0.41$, $fr_4 = 0.571$, $fr_5 = 0.27$, $fr_6 = 0.625$, $fr_7 = 0.435$, $fr_8 = 0.488$, $fr_9 = 0.573$, and $fr_{10} = 0.588$.

Now we construct the BST of the given requirements on the basis of the rv that we have obtained in the previous step (see Fig. 6). Then we traverse the BST in IN-ORDER to get the following order:

$fr_5 < fr_3 < fr_7 < fr_8 < fr_2 < fr_4 < fr_9 < fr_1 < fr_{10} < fr_6$.

In our case study, fr_6 has the highest priority and fr_5 has the lowest priority. Among all the modules, student's

module has the highest priority, so this module would be implemented first.

5 Conclusion

The problem of RP in goal oriented requirements elicitation process is a MCDM problem as it includes several stakeholder's participation. This paper presents a method for PRFGORE process. In order to strengthen the goal oriented requirements elicitation process; and to make decision when the information is fuzzy and imprecise, in proposed approach, we used α -level weighted F-preference relation and extent fuzzy AHP in group decision making process. To get the prioritized list of requirements we used binary tree sort method. To simply show how proposed approach works, a numerical example is shown to illustrate the fuzzy group decision making approach in goal oriented requirements elicitation process.

In our example, we assumed that there are ten FRs, three criteria, i.e., security, reliability, and performance, for the prioritization of requirements and; ten stakeholders' and five DM are participating to prioritize the requirements. On the basis of our analysis, we identify that fr_6 is the most important requirements and it would be implemented first; and fr_5 would be implemented in the last. However, the method discussed in this paper can be further exploited by considering more sub-goals/requirements and criteria as well as much larger group of stakeholders in group decision making process. The future research agenda can be listed as follows:

1. To propose a fuzzy attributed goal oriented requirements elicitation and analysis (FAGOREA) method by using fuzzy contribution values and fuzzy preference matrix.
2. To develop a tool to support the decision making processes of FAGOREA.

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